GATE ME 90 Days Challenge

Day 47/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Applied Thermodynamics	Power Cycles	57 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on reciprocating air compressors (clearance, volumetric efficiency, multi-staging and intercooling), the Rankine cycle with superheat, reheat and regenerative feed heating, and the Brayton gas turbine cycle with component efficiencies, regeneration, intercooling and combined cycles. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The Carnot cycle is not used as the practical model for steam power plants mainly because

(A) its efficiency is lower than that of the Rankine cycle

(B) it would require compressing a wet (two-phase) mixture in the pump and expanding very wet steam in the turbine

(C) it needs a condenser

(D) it cannot use superheated steam at all

Q2 NAT

In a simple Rankine cycle the steam enters the turbine at 8 MPa with h_1 = 3230 kJ/kg and leaves the isentropic turbine at 10 kPa with h_2 = 2140 kJ/kg. Saturated liquid leaves the condenser with h_3 = 192 kJ/kg and v = 0.00101 m³/kg. The cycle efficiency (in %, round off to two decimal places) is _____

Q3 MCQ

In a Rankine cycle the boiler pressure is raised, keeping the maximum steam temperature and the condenser pressure unchanged. Then

(A) the efficiency falls and the exhaust becomes drier

(B) both the efficiency and the exhaust dryness increase

(C) the efficiency rises but the steam at the turbine exhaust becomes wetter

(D) neither changes

Q4 NAT

Air (R = 0.287 kJ/kg·K) at 1 bar and 300 K is compressed to 9 bar in a two-stage reciprocating compressor with perfect intercooling and the optimum intermediate pressure. Compression follows $pV^{1.3}$ = constant in both stages. The minimum total work per kg of air (in kJ/kg, round off to one decimal place) is _____

Q5 MCQ

The volumetric efficiency of a single-stage reciprocating air compressor with a fixed clearance volume

(A) increases as the delivery pressure increases
(B) is independent of the pressure ratio
(C) is 100 % whenever the compression is isothermal
(D) decreases as the pressure ratio increases

Q6 NAT

Air (R = 0.287 kJ/kg·K, γ = 1.4) at 300 K is compressed from 1 bar to 6 bar in a steady-flow compressor. The isothermal efficiency, defined as (isothermal work)/(isentropic work) for the same pressure ratio (in %, round off to two decimal places), is _____

Q7 MCQ

The thermal efficiency of an ideal (air-standard) Brayton cycle depends on

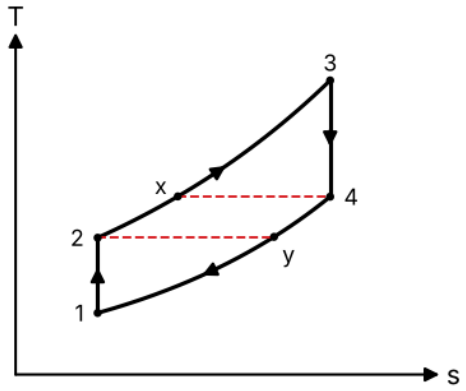
(A) the pressure ratio only (for a given γ)
(B) the maximum temperature only
(C) both the maximum and minimum temperatures
(D) the mass flow rate of air

Q8 NAT

An ideal Brayton cycle works on air (c_p = 1.005 kJ/kg·K, γ = 1.4) with a compressor inlet temperature of 300 K, a pressure ratio of 8 and a turbine inlet temperature of 1300 K. The back work ratio (compressor work/turbine work, round off to three decimal places) is _____

Q9 MCQ

The T–s diagram shows an ideal Brayton cycle 1–2–3–4 with an ideal regenerator, in which the compressed air (2 → x) is heated by the turbine exhaust (4 → y). For fixed T_1 and T_3, the thermal efficiency of this cycle



(A) increases as the pressure ratio increases
(B) decreases as the pressure ratio increases
(C) is independent of the pressure ratio
(D) equals the Carnot efficiency at every pressure ratio

Q10 NAT

A gas turbine plant works on air (c_p = 1.005 kJ/kg·K, γ = 1.4). Air enters the compressor at 290 K; the pressure ratio is 6 and the turbine inlet temperature is 1100 K. The isentropic efficiencies of the compressor and turbine are 0.82 and 0.88. The thermal efficiency of the plant (in %, round off to two decimal places) is _____

Q11 MCQ

Compared with a closed feed-water heater, an open (direct-contact) feed-water heater

(A) needs a separate pump after it to raise the feed water to the next pressure

(B) cannot be used for deaeration

(C) keeps the bled steam and feed water separate

(D) is always more expensive

Q12 NAT

A steam power plant has a net work output of 1125 kJ/kg of steam and receives 3000 kJ/kg of heat in the boiler. Its heat rate (in kJ/kWh) is _____

Q13 NAT

Air (R = 0.287 kJ/kg·K) at 1 bar and 300 K is compressed to 27 bar in a three-stage reciprocating compressor with perfect intercooling and optimum stage pressures. Compression follows $pV^{1.25}$ = constant. The minimum total work per kg of air (in kJ/kg, round off to one decimal place) is _____

Q14 MCQ

In a combined gas turbine–steam turbine power plant,

(A) the steam cycle is the topping cycle and the gas turbine the bottoming cycle

(B) the gas turbine is the topping cycle and its exhaust raises steam for the bottoming Rankine cycle

(C) the two cycles share the same working fluid

(D) the overall efficiency is the product of the two cycle efficiencies

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Q15 MCQ

A gas turbine cycle uses many stages of intercooling and of reheating together with an ideal regenerator. As the number of stages tends to infinity, its efficiency approaches that of the

(A) Otto cycle

(B) Diesel cycle

(C) Rankine cycle

(D) Ericsson cycle, which equals the Carnot efficiency between the same temperatures

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Answer Key

Q	Type	Answer
Q1	MCQ	(B) it would require compressing a wet (two-phase) mixture in the pump and expanding very wet steam in the turbine
Q2	NAT	35.71 (accept 35.50 to 35.90)
Q3	MCQ	(C) the efficiency rises but the steam at the turbine exhaust becomes wetter
Q4	NAT	215.3 (accept 213.6 to 217.0)
Q5	MCQ	(D) decreases as the pressure ratio increases
Q6	NAT	76.58 (accept 76.00 to 77.10)
Q7	MCQ	(A) the pressure ratio only (for a given γ)
Q8	NAT	0.418 (accept 0.414 to 0.422)
Q9	MCQ	(B) decreases as the pressure ratio increases
Q10	NAT	26.40 (accept 26.20 to 26.60)
Q11	MCQ	(A) needs a separate pump after it to raise the feed water to the next pressure
Q12	NAT	9600
Q13	NAT	317.4 (accept 315.0 to 319.8)
Q14	MCQ	(B) the gas turbine is the topping cycle and its exhaust raises steam for the bottoming Rankine cycle
Q15	MCQ	(D) Ericsson cycle, which equals the Carnot efficiency between the same temperatures

Detailed Solutions

Q1 Answer: (B) it would require compressing a wet (two-phase) mixture in the pump and expanding very wet steam in the turbine

Concept: A Carnot vapour cycle adds and rejects heat isothermally inside the wet region.

Its compression would start from a wet mixture: a two-phase compressor is impractical and absorbs large work.

Its expansion ends with very wet steam, which erodes turbine blades. The Rankine cycle condenses fully and pumps liquid instead: **(B)**.

Exam tip: Carnot efficiency is the upper limit; the Rankine cycle trades a little efficiency for practicality.

Q2 **Answer: 35.71 (accept 35.50 to 35.90)**

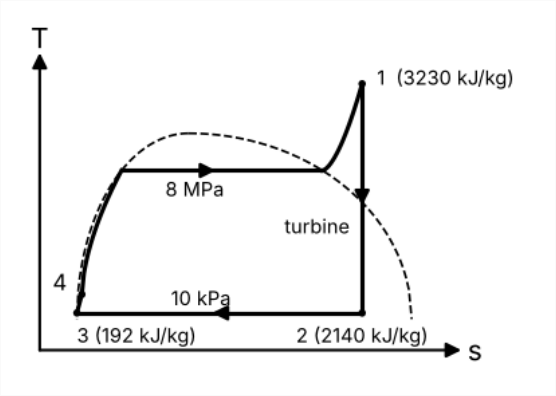
Concept: $\eta = (w_T - w_P)/q_{in}$, with the pump work $w_P = v(p_{boiler} - p_{cond})$.

$w_P = 0.00101 \times (8000 - 10) = 8.07$ kJ/kg; $h_4 = 192 + 8.07 = 200.07$ kJ/kg

$w_T = 3230 - 2140 = 1090$ kJ/kg; $w_{net} = 1081.93$ kJ/kg

$q_{in} = 3230 - 200.07 = 3029.93$ kJ/kg; $\eta = 1081.93/3029.93 =$ **35.71 %**

Exam tip: Here the pump work is only 0.7 % of the turbine work; neglecting it changes η by about 0.1 %.



Q3 **Answer: (C) the efficiency rises but the steam at the turbine exhaust becomes wetter**

Concept: Efficiency depends on the mean temperature of heat addition.

Higher boiler pressure raises the saturation temperature, so the mean temperature of heat addition rises and η increases.

But at the same maximum temperature the expansion line shifts to lower entropy, ending deeper in the wet region: **wetter exhaust (C)**.

Exam tip: This is exactly why reheat is used with high boiler pressures.

Q4 **Answer: 215.3 (accept 213.6 to 217.0)**

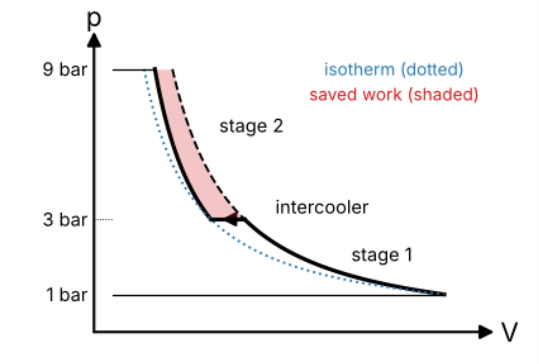
Concept: With perfect intercooling (back to T_1) and $p_2 = \sqrt{(p_1 p_3)}$, both stages do equal work: $W = 2[n/(n-1)]RT_1[(p_3/p_1)^{(n-1)/(2n)} - 1]$.

$p_2 = \sqrt{9} = 3$ bar; each stage has a ratio of 3; $(n-1)/n = 0.2308$

$3^{0.2308} = 1.2885$

$W = 2 \times 4.333 \times 0.287 \times 300 \times 0.2885 =$ **215.3 kJ/kg**

Exam tip: Single-stage compression to 9 bar would need $4.333 \times 86.1 \times (9^{0.2308} - 1) = 246.4$ kJ/kg: intercooling saves about 13 %.



Q5 **Answer: (D) decreases as the pressure ratio increases**

Concept: $\eta_v = 1 + C - C(p_2/p_1)^{1/n}$.

The clearance air re-expands to a larger volume when the pressure ratio is larger, so less fresh air is drawn in.

Hence η_v **decreases with pressure ratio (D)** and becomes zero at the limiting ratio $(1 + 1/C)^n$.

Exam tip: This is the second reason (besides work saving) for multi-stage compression at high pressure ratios.

Q6 **Answer: 76.58 (accept 76.00 to 77.10)**

Concept: Isothermal work = $RT_1 \ln r$; isentropic work = $[\gamma/(\gamma - 1)]RT_1(r^{(\gamma-1)/\gamma} - 1)$.

Isothermal: $0.287 \times 300 \times \ln 6 = 86.1 \times 1.7918 = 154.27$ kJ/kg

Isentropic: $3.5 \times 86.1 \times (6^{0.2857} - 1) = 301.35 \times 0.6685 = 201.45$ kJ/kg

Ratio = 154.27/201.45 = **76.58 %**

Exam tip: RT_1 cancels, so the ratio depends only on the pressure ratio and γ.

Q7 **Answer: (A) the pressure ratio only (for a given γ)**

Concept: $\eta = 1 - 1/r_p^{(\gamma-1)/\gamma} = 1 - T_1/T_2$.

Neither T_{max} nor the flow rate appears: **(A)**.

The turbine inlet temperature does, however, decide the net work per kg (and the optimum pressure ratio).

Exam tip: With component inefficiencies, η also depends on T_3/T_1: a hotter turbine inlet then helps efficiency too.

Q8 **Answer: 0.418 (accept 0.414 to 0.422)**

Concept: Back work ratio = $w_C/w_T = (T_2 - T_1)/(T_3 - T_4)$.

$8^{0.2857} = 1.8114$; $T_2 = 300 \times 1.8114 = 543.4$ K; $T_4 = 1300/1.8114 = 717.7$ K

BWR = $(543.4 - 300)/(1300 - 717.7) = 243.4/582.3$ = **0.418**

Shortcut: since $T_2/T_1 = T_3/T_4$, BWR = $T_2/T_3 = 543.4/1300$.

Exam tip: Gas turbines spend 40–60 % of turbine work on the compressor; a steam plant spends about 1 %.

Q9 **Answer: (B) decreases as the pressure ratio increases**

Concept: For an ideal regenerative Brayton cycle, $\eta = 1 - (T_1/T_3)r_p^{(\gamma-1)/\gamma}$.

As r_p rises, T_4 falls towards T_2 and less heat can be recovered, so η **decreases (B)**.

At $r_p \to 1$ it tends to the Carnot value $1 - T_1/T_3$ (but with zero net work).

Exam tip: This is the opposite trend to the simple Brayton cycle, whose efficiency rises with r_p.

Q10 **Answer: 26.40 (accept 26.20 to 26.60)**

Concept: Find the actual exit temperatures from the isentropic efficiencies, then $\eta = w_{net}/q_{in}$.

$6^{0.2857} = 1.6685$. $T_{2s} = 483.9$ K; $T_2 = 290 + 193.9/0.82 = 526.4$ K

$T_{4s} = 1100/1.6685 = 659.3$ K; $T_4 = 1100 - 0.88 \times 440.7 = 712.2$ K

$w_{net}/c_p = (1100 - 712.2) - (526.4 - 290) = 151.4$ K; $q_{in}/c_p = 1100 - 526.4 = 573.6$ K

$\eta = 151.4/573.6 =$ **26.40 %** (ideal cycle: 40.07 %)

Exam tip: Component losses hit gas turbines hard because the net work is a small difference of two large works.

Q11 **Answer: (A) needs a separate pump after it to raise the feed water to the next pressure**

Concept: In an open heater the bled steam mixes with the feed water at the bleed pressure.

The mixture leaves as saturated liquid at that (lower) pressure, so **a pump is required after each open heater (A)**.

Open heaters are cheap, give good heat transfer and also remove dissolved gases (deaerator).

Exam tip: Closed heaters keep the streams apart (a shell-and-tube exchanger), so only one feed pump is needed.

Q12 **Answer: 9600**

Concept: Heat rate = heat supplied per kWh of net work = $3600/\eta$ kJ/kWh.

$\eta = 1125/3000 = 0.375$

Heat rate = $3600/0.375 =$ **9600 kJ/kWh**

Exam tip: The lower the heat rate, the better the plant; the steam rate $3600/w_{net} = 3.2$ kg/kWh is a different measure.

Q13 **Answer: 317.4 (accept 315.0 to 319.8)**

Concept: For N stages with perfect intercooling the optimum stage ratio is $(p_{final}/p_1)^{1/N}$ and $W = N[n/(n-1)]RT_1[r_s^{(n-1)/n} - 1]$.

$r_s = 27^{1/3} = 3$ (stage pressures 1, 3, 9, 27 bar); $(n-1)/n = 0.2$

$3^{0.2} = 1.2457$

$W = 3 \times 5 \times 0.287 \times 300 \times 0.2457 =$ **317.4 kJ/kg**

Exam tip: Each stage does equal work and has equal delivery temperature ($300 \times 1.2457 = 373.7$ K).

Q14 **Answer: (B) the gas turbine is the topping cycle and its exhaust raises steam for the bottoming Rankine cycle**

Concept: The gas turbine works at high temperature and rejects heat at 450–600 °C, ideal for raising steam.

So the gas turbine is the **topping cycle (B)** and a heat-recovery steam generator feeds the Rankine bottoming cycle.

$\eta_{overall} = \eta_1 + \eta_2 - \eta_1\eta_2$, which exceeds either efficiency.

Exam tip: Modern combined-cycle plants reach about 60 % efficiency.

Q15 Answer: (D) Ericsson cycle, which equals the Carnot efficiency between the same temperatures

Concept: Infinite intercooling makes compression isothermal at T_{min}; infinite reheating makes expansion isothermal at T_{max}.

With an ideal regenerator handling the constant-pressure processes, all external heat is exchanged at T_{max} and T_{min}.

That is the **Ericsson cycle (D)**, whose efficiency is $1 - T_{min}/T_{max}$.

Exam tip: In practice two or three stages give most of the gain.

What next?

Day 47 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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